

Pathology

“Digressions, objections, delight in mockery, carefree mistrust are signs of health; everything unconditional belongs in pathology.”

—Friedrich Nietzsche

“You cannot separate passion from pathology any more than you can separate a person’s spirit from his body.”

—Richard Selzer

The fundamental principles of pathology are key to understanding diseases in all organ systems. Major topics such as inflammation and neoplasia appear frequently in questions across different organ systems, and such topics are definitely high yield. For example, the concepts of cell injury and inflammation are key to understanding the inflammatory response that follows myocardial infarction, a very common subject of board questions. Similarly, a familiarity with the early cellular changes that culminate in the development of neoplasias—for example, esophageal or colon cancer—is critical. Finally, make sure you recognize the major tumor-associated genes and are comfortable with key cancer concepts such as tumor staging and metastasis.

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▶ PATHOLOGY—INFLAMMATION

Apoptosis

ATP-dependent programmed cell death.

Intrinsic and extrinsic pathways; both pathways activate caspases (cytosolic proteases) → cellular breakdown including cell shrinkage, chromatin condensation, membrane blebbing, and formation of apoptotic bodies, which are then phagocytosed.

Characterized by deeply eosinophilic cytoplasm and basophilic nucleus, pyknosis (nuclear shrinkage), and karyorrhexis (fragmentation caused by endonuclease-mediated cleavage).

Cell membrane typically remains intact without significant inflammation (unlike necrosis).

DNA laddering (fragments in multiples of 180 bp) is a sensitive indicator of apoptosis.

Intrinsic (mitochondrial) pathway

Involved in tissue remodeling in embryogenesis. Occurs when a regulating factor is withdrawn from a proliferating cell population (eg, ↓ IL-2 after a completed immunologic reaction → apoptosis of proliferating effector cells). Also occurs after exposure to injurious stimuli (eg, radiation, toxins, hypoxia).

Regulated by Bcl-2 family of proteins. BAX and BAK are proapoptotic, while Bcl-2 and Bcl-x are antiapoptotic.

Bcl-2 keeps the mitochondrial outer membrane impermeable and therefore prevents cytochrome c release from the inner mitochondrial matrix. Bcl-2 overexpression (eg, follicular lymphoma t[14;18]) → ↓ caspase activation → tumorigenesis.

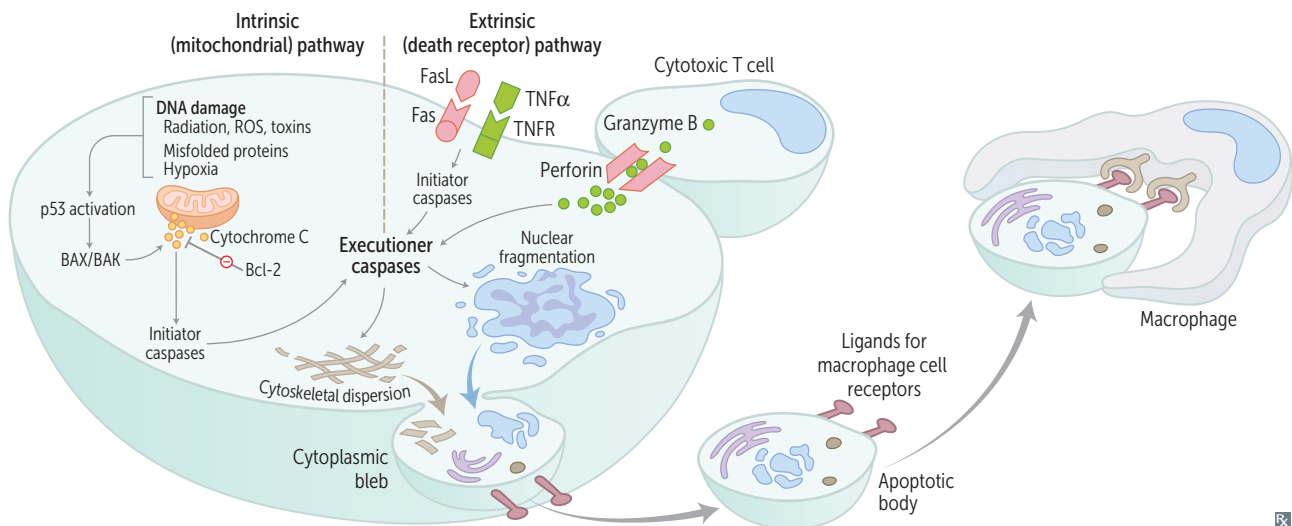
Extrinsic (death receptor) pathway

2 pathways:

- Ligand receptor interactions (FasL binding to Fas [CD95] or TNF-α binding to its receptor)
- Immune cell (cytotoxic T-cell release of perforin and granzyme B)

Fas-FasL interaction is necessary in thymic medullary negative selection. Mutations in Fas ↑ numbers of circulating self-reacting lymphocytes due to failure of clonal deletion.

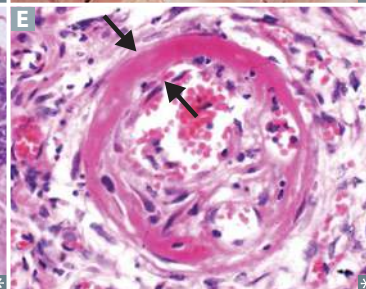
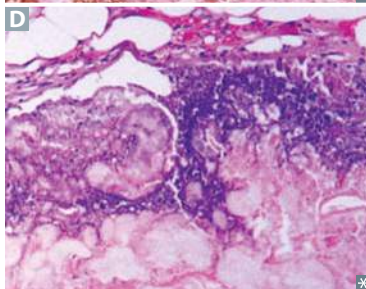
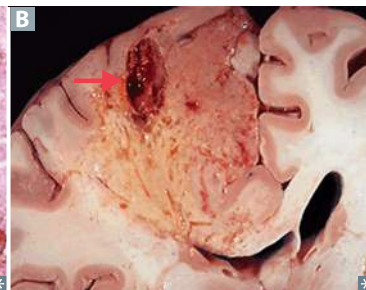
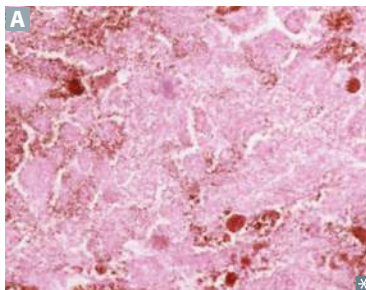
Defective Fas-FasL interactions cause autoimmune lymphoproliferative syndrome.



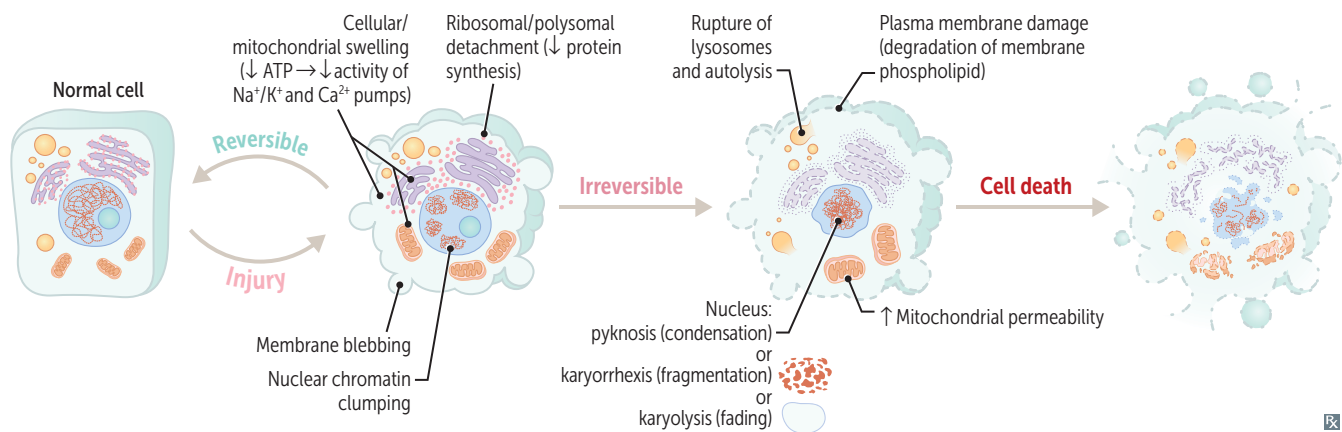
Necrosis

Enzymatic degradation and protein denaturation of cell due to exogenous injury → intracellular components leak. Inflammatory process (unlike apoptosis).

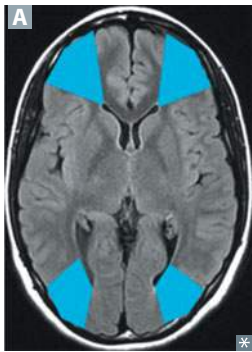
TYPE	SEEN IN	DUE TO	HISTOLOGY
Coagulative	Ischemia/infarcts in most tissues (except brain)	Ischemia or infarction; proteins denature, then enzymatic degradation	Cell outlines preserved but nuclei disappear; ↑ cytoplasmic binding of eosin dyes A
Liquefactive	Bacterial abscesses, brain infarcts	Neutrophils release lysosomal enzymes that digest the tissue B ; enzymatic degradation first, then proteins denature	Early: cellular debris and macrophages Late: cystic spaces and cavitation (brain) Neutrophils and cell debris seen with bacterial infection
Caseous	TB, systemic fungi (eg, <i>Histoplasma capsulatum</i>), <i>Nocardia</i>	Macrophages wall off the infecting microorganism → granular debris C	Fragmented cells and debris surrounded by lymphocytes and macrophages
Fat	Enzymatic: acute pancreatitis (saponification of peripancreatic fat) Nonenzymatic: traumatic (eg, injury to breast tissue)	Damaged cells release lipase to break down triglycerides, liberating fatty acids to bind calcium → saponification	Outlines of dead fat cells without peripheral nuclei; saponification of fat (combined with Ca^{2+}) appears dark blue on H&E stain D
Fibrinoid	Immune reactions in vessels (eg, polyarteritis nodosa), preeclampsia, malignant hypertension	Immune complexes combine with fibrin → vessel wall damage (type III hypersensitivity reaction)	Vessel walls are thick and pink E
Gangrenous	Distal extremity and GI tract, after chronic ischemia	Dry: ischemia F Wet: superinfection	Coagulative Liquefactive superimposed on coagulative



Cell injury



Ischemia



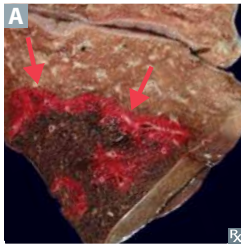
Inadequate blood supply to meet demand.

Regions most vulnerable to hypoxia/ischemia and subsequent infarction:

ORGAN	REGION
Brain	ACA/MCA/PCA boundary areas ^{a,b}
Heart	Subendocardium (LV)
Kidney	Straight segment of proximal tubule (medulla) Thick ascending limb (medulla)
Liver	Area around central vein (zone III)
Colon	Splenic flexure, ^a rectum ^a

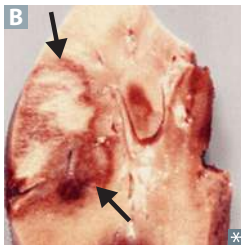
^aWatershed areas (border zones) receive blood supply from most distal branches of 2 arteries with limited collateral vascularity. These areas are susceptible to ischemia from hypoperfusion (eg, ACA/MCA [anterior] and MCA/PCA [posterior] watershed areas in blue **A**).

^bNeurons most vulnerable to hypoxic-ischemic insults include Purkinje cells of the cerebellum and pyramidal cells of the hippocampus and neocortex (zones 3, 5, 6).

Types of infarcts**Red infarct**

Red (hemorrhagic) infarcts **A** occur in venous occlusion and tissues with multiple blood supplies, such as liver, lung, intestine, testes; reperfusion (eg, after angioplasty). Reperfusion injury is due to damage by free radicals.

Red = reperfusion.

Pale infarct

Pale (anemic) infarcts **B** occur in solid organs with a single (end-arterial) blood supply, such as heart, kidney, and spleen.

Inflammation

Characterized by *rubor* (redness), *dolor* (pain), *calor* (heat), *tumor* (swelling), and *functio laesa* (loss of function).

Vascular component

↑ vascular permeability, vasodilation, endothelial injury.

Cellular component

Neutrophils extravasate from circulation to injured tissue to participate in inflammation through phagocytosis, degranulation, and inflammatory mediator release.

Acute

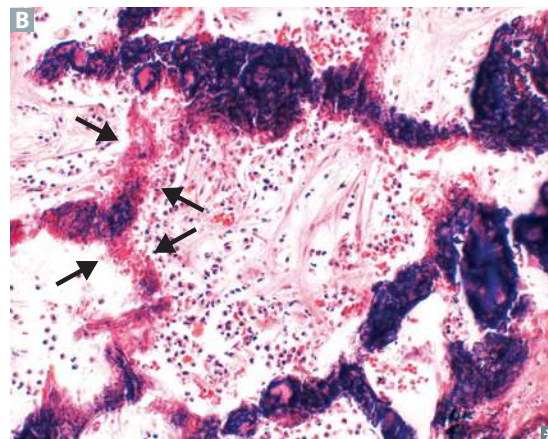
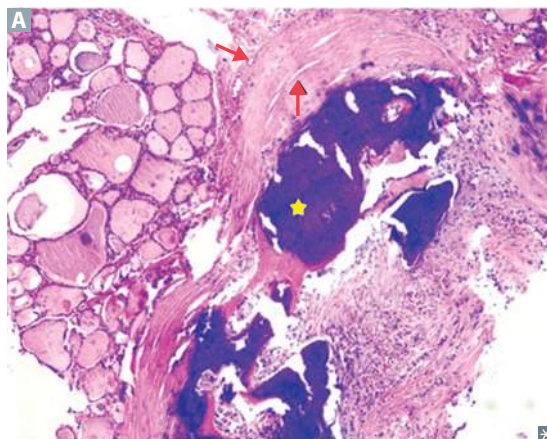
Neutrophil, eosinophil, antibody (pre-existing), mast cell, and basophil mediated. Acute inflammation is rapid onset (seconds to minutes) and of short duration (minutes to days). Outcomes include complete resolution, abscess formation, or progression to chronic inflammation.

Chronic

Mononuclear cell (monocytes/macrophages, lymphocytes, plasma cells) and fibroblast mediated. Characterized by persistent destruction and repair. Associated with blood vessel proliferation, fibrosis. Granuloma: nodular collections of epithelioid macrophages and giant cells. Outcomes include scarring, amyloidosis, and neoplastic transformation.

Types of calcification

Types of calcification	Dystrophic calcification	Metastatic calcification
Ca²⁺ DEPOSITION	In abnormal tissues	In normal tissues
EXTENT	Tends to be localized (eg, calcific aortic stenosis) A shows dystrophic calcification (yellow star), and thick fibrotic wall (red arrows)	Widespread (ie, diffuse, metastatic) B shows metastatic calcifications of alveolar walls in acute pneumonitis (arrows)
ASSOCIATED CONDITIONS	TB (lung and pericardium) and other granulomatous infections; liquefactive necrosis of chronic abscesses; fat necrosis; infarcts; thrombi; schistosomiasis; congenital CMV, toxoplasmosis, rubella; psammoma bodies; CREST syndrome	Predominantly in interstitial tissues of kidney, lung, and gastric mucosa (these tissues lose acid quickly; ↑ pH favors Ca ²⁺ deposition) Nephrocalcinosis of collecting ducts may lead to nephrogenic diabetes insipidus and renal failure
ETIOLOGY	2° to injury or necrosis	2° to hypercalcemia (eg, 1° hyperparathyroidism, sarcoidosis, hypervitaminosis D) or high calcium-phosphate product levels (eg, chronic renal failure with 2° hyperparathyroidism, long-term dialysis, calciphylaxis, multiple myeloma)
SERUM CA²⁺ LEVELS	Patients are usually normocalcemic	Patients are usually not normocalcemic

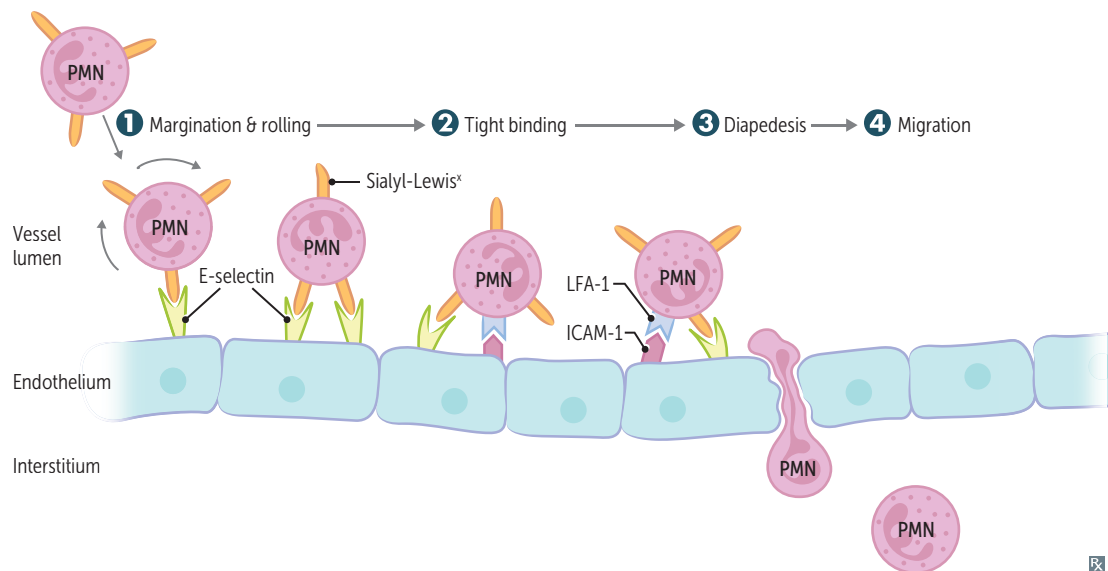


Leukocyte extravasation

Extravasation predominantly occurs at postcapillary venules.

WBCs exit from blood vessels at sites of tissue injury and inflammation in 4 steps:

STEP	VASCULATURE/STROMA	LEUKOCYTE
1 Margination and rolling— defective in leukocyte adhesion deficiency type 2 (↓ Sialyl-Lewis ^X)	E-selectin (upregulated by TNF and IL-1)	Sialyl-Lewis ^X
	P-selectin (released from Weibel-Palade bodies)	Sialyl-Lewis ^X
	GlyCAM-1, CD34	L-selectin
2 Tight binding (adhesion)— defective in leukocyte adhesion deficiency type 1 (↓ CD18 integrin subunit)	ICAM-1 (CD54)	CD11/18 integrins (LFA-1, Mac-1)
	VCAM-1 (CD106)	VLA-4 integrin
3 Diapedesis—WBC travels between endothelial cells and exits blood vessel	PECAM-1 (CD31)	PECAM-1 (CD31)
4 Migration—WBC travels through interstitium to site of injury or infection guided by chemotactic signals	Chemotactic products released in response to bacteria: C5a, IL-8, LTB ₄ , kallikrein, platelet-activating factor	Various



Free radical injury

Free radicals damage cells via membrane lipid peroxidation, protein modification, and DNA breakage.

Initiated via radiation exposure (eg, cancer therapy), metabolism of drugs (phase I), redox reactions, nitric oxide (eg, inflammation), transition metals, WBC (eg, neutrophils, macrophages) oxidative burst.

Free radicals can be eliminated by scavenging enzymes (eg, catalase, superoxide dismutase, glutathione peroxidase), spontaneous decay, antioxidants (eg, vitamins A, C, E), and certain metal carrier proteins (eg, transferrin, ceruloplasmin).

Examples:

- Oxygen toxicity: retinopathy of prematurity (abnormal vascularization), bronchopulmonary dysplasia, reperfusion injury after thrombolytic therapy
- Drug/chemical toxicity: carbon tetrachloride and acetaminophen overdose (hepatotoxicity)
- Metal storage diseases: hemochromatosis (iron) and Wilson disease (copper)

Scar formation

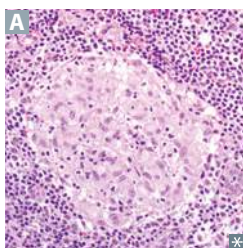
70–80% of tensile strength regained at 3 months; minimal additional tensile strength will be regained afterward.

SCAR TYPE	Hypertrophic A	Keloid B
COLLAGEN SYNTHESIS	↑ (type III collagen)	↑↑↑ (disorganized types I and III collagen)
COLLAGEN ORGANIZATION	Parallel	Disorganized
EXTENT OF SCAR	Confined to borders of original wound	Extends beyond borders of original wound with “claw-like” projections typically on earlobes, face, upper extremities
RECURRENCE	Infrequent	Frequent
PREDISPOSITION	None	↑ incidence in ethnic groups with darker skin



Wound healing

Tissue mediators	MEDIATOR	ROLE
	PDGF	Secreted by activated platelets and macrophages Induces vascular remodeling and smooth muscle cell migration Stimulates fibroblast growth for collagen synthesis
	FGF	Stimulates angiogenesis
	EGF	Stimulates cell growth via tyrosine kinases (eg, EGFR/ <i>ErbB1</i>)
	TGF- β	Angiogenesis, fibrosis
	Metalloproteinases	Tissue remodeling
	VEGF	Stimulates angiogenesis
PHASE OF WOUND HEALING	EFFECTOR CELLS	CHARACTERISTICS
Inflammatory (up to 3 days after wound)	Platelets, neutrophils, macrophages	Clot formation, $\uparrow$ vessel permeability and neutrophil migration into tissue; macrophages clear debris 2 days later
Proliferative (day 3–weeks after wound)	Fibroblasts, myofibroblasts, endothelial cells, keratinocytes, macrophages	Deposition of granulation tissue and type III collagen, angiogenesis, epithelial cell proliferation, dissolution of clot, and wound contraction (mediated by myofibroblasts) Delayed wound healing in vitamin C deficiency and copper deficiency
Remodeling (1 week–6+ months after wound)	Fibroblasts	Type III collagen replaced by type I collagen, $\uparrow$ tensile strength of tissue Delayed wound healing in zinc deficiency

Granulomatous diseases**Bacterial:**

- *Mycobacteria* (tuberculosis, leprosy)
- *Bartonella henselae* (cat scratch disease)
- *Listeria monocytogenes* (granulomatosis infantiseptica)
- *Treponema pallidum* (3° syphilis)

Fungal: endemic mycoses (eg, histoplasmosis)

Parasitic: schistosomiasis

Chronic granulomatous disease

Autoinflammatory:

- Sarcoidosis
- Crohn disease
- Primary biliary cirrhosis
- Subacute (de Quervain/granulomatous) thyroiditis
- Granulomatosis with polyangiitis (Wegener)
- Eosinophilic granulomatosis with polyangiitis (Churg-Strauss)
- Giant cell (temporal) arteritis
- Takayasu arteritis

Foreign material: berylliosis, talcosis, hypersensitivity pneumonitis

Granulomas are composed of epithelioid cells (macrophages with abundant pink cytoplasm) with surrounding multinucleated giant cells and lymphocytes. Th₁ cells secrete IFN- γ , activating macrophages. TNF- α from macrophages induces and maintains granuloma formation. Anti-TNF drugs can, as a side effect, cause sequestering granulomas to break down, leading to disseminated disease. Always test for latent TB before starting anti-TNF therapy.

Associated with hypercalcemia due to calcitriol (1,25-[OH]₂ vitamin D₃) production.

Caseating necrosis is more common with an infectious etiology (eg, TB). Diagnosis of sarcoidosis requires noncaseating granulomas **A** on biopsy.

Exudate vs transudate**Exudate**

Cellular (cloudy)

↑ protein (> 2.9 g/dL)

↑ LDH (vs serum)

Due to:

- Lymphatic obstruction (chylous)
- Inflammation/infection
- Malignancy

Transudate

Hypocellular (clear)

↓ protein (< 2.5 g/dL)

↓ LDH (vs serum)

Due to:

- ↑ hydrostatic pressure (eg, HF, Na⁺ retention)
- ↓ oncotic pressure (eg, cirrhosis, nephrotic syndrome)

Light criteria

Diagnostic analysis comparing serum and pleural fluid protein and LDH levels.

Pleural effusion is exudative if ≥ 1 of the following criteria is met:

- Pleural effusion protein/serum protein ratio > 0.5
- Pleural effusion LDH/serum LDH ratio > 0.6
- Pleural effusion LDH > ⅔ of the upper limit of normal for serum LDH

Erythrocyte sedimentation rate

Products of inflammation (eg, fibrinogen) coat RBCs and cause aggregation. The denser RBC aggregates fall at a faster rate within a pipette tube. Often co-tested with CRP levels.

↑ ESR

Most anemias

Infections

Inflammation (eg, giant cell [temporal] arteritis, polymyalgia rheumatica)

Cancer (eg, metastases, multiple myeloma)

Renal disease (end-stage or nephrotic syndrome)

Pregnancy

↓ ESR

Sickle cell anemia (altered shape)

Polycythemia (↑ RBCs “dilute” aggregation factors)

HF

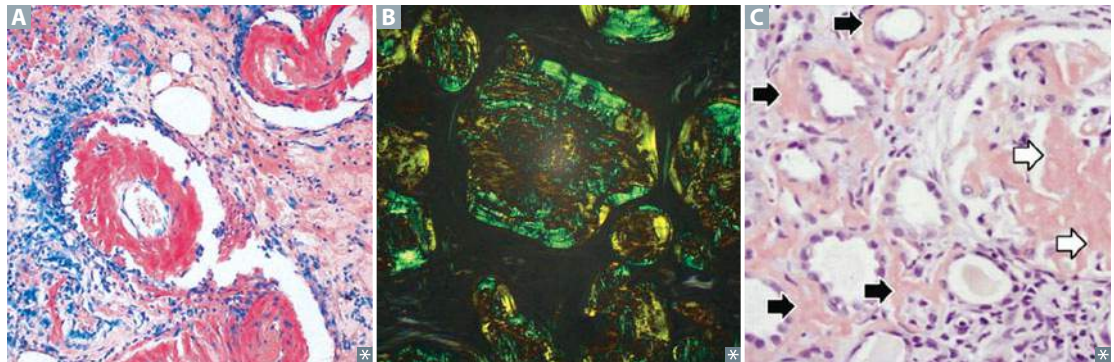
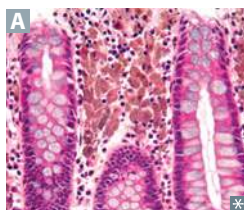
Microcytosis

Hypofibrinogenemia

Amyloidosis

Abnormal aggregation of proteins (or their fragments) into β -pleated linear sheets $\rightarrow$ insoluble fibrils $\rightarrow$ cellular damage and apoptosis. Amyloid deposits visualized by Congo red stain **A**, polarized light (apple green birefringence) **B**, and H&E stain (**C** shows deposits in glomerular mesangial areas [white arrows], tubular basement membranes [black arrows]).

COMMON TYPES	DESCRIPTION
AL (primary)	Due to deposition of proteins from Ig L ight chains. Can occur as a plasma cell disorder or associated with multiple myeloma. Often affects multiple organ systems, including renal (nephrotic syndrome), cardiac (restrictive cardiomyopathy, arrhythmia), hematologic (easy bruising, splenomegaly), GI (hepatomegaly), and neurologic (neuropathy).
AA (secondary)	Seen with chronic inflammatory conditions such as rheumatoid arthritis, IBD, spondyloarthropathy, familial Mediterranean fever, protracted infection. Fibrils composed of serum A myloid A . Often multisystem like AL amyloidosis.
Dialysis-related	Fibrils composed of β_2 -microglobulin in patients with ESRD and/or on long-term dialysis. May present as carpal tunnel syndrome.
Heritable	Heterogeneous group of disorders, including familial amyloid polyneuropathies due to transthyretin gene mutation.
Age-related (senile) systemic	Due to deposition of normal (wild-type) transthyretin (TTR) predominantly in cardiac ventricles. Slower progression of cardiac dysfunction relative to AL amyloidosis.
Organ-specific	Amyloid deposition localized to a single organ. Most important form is amyloidosis in Alzheimer disease due to deposition of β -amyloid protein cleaved from amyloid precursor protein (APP). Islet amyloid polypeptide (IAPP) is commonly seen in diabetes mellitus type 2 and is caused by deposition of amylin in pancreatic islets. Isolated atrial amyloidosis due to atrial natriuretic peptide is common in normal aging and can predispose to increased risk of atrial fibrillation. Amyloid deposition to ventricular endomyocardium in restrictive cardiomyopathy. Calcitonin deposition in tumor cells in medullary carcinoma of the thyroid.

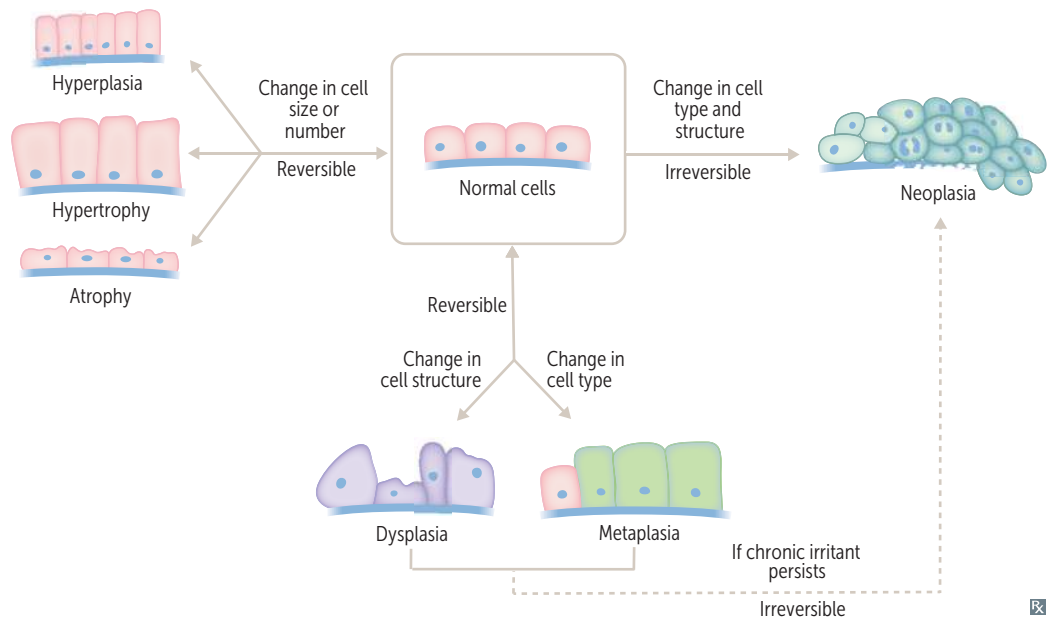
**Lipofuscin**

A yellow-brown “wear and tear” pigment **A** associated with normal aging. Formed by oxidation and polymerization of autophagocytosed organellar membranes. Autopsy of elderly person will reveal deposits in heart, colon, liver, kidney, eye, and other organs.

▶ PATHOLOGY—NEOPLASIA

Cellular changes

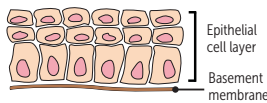
Hyperplasia	↑ in number of cells. May be a risk factor for future malignancy (eg, endometrial hyperplasia) but not considered premalignant.
Hypertrophy	↑ in size of cells.
Atrophy	↓ in tissue mass due to ↓ in size and/or number of cells. Causes include disuse, denervation, loss of blood supply, loss of hormonal stimulation, poor nutrition.
Dysplasia	Disordered, non-neoplastic cell growth. Term used only with epithelial cells. Mild dysplasia is usually reversible; severe dysplasia usually progresses to carcinoma in situ.
Metaplasia	Replacement of one cell type by another. Usually due to exposure to an irritant, such as gastric acid or cigarette smoke. Reversible if the irritant is removed but may undergo malignant transformation with persistent insult (eg, Barrett esophagus → esophageal adenocarcinoma).
Neoplasia	Uncontrolled, clonal proliferation of cells. Can be benign or malignant.
Anaplasia	Complete lack of differentiation of cells in a malignant neoplasm.
Differentiation	The degree to which a malignant tumor resembles its tissue of origin: - Well-differentiated tumors (often less aggressive) closely resemble their tissue of origin. - Poorly differentiated tumors (often more aggressive) look almost nothing like their tissue of origin.



Neoplastic progression

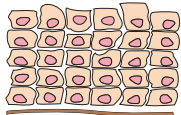
Hallmarks of cancer: evasion of apoptosis, growth signal self-sufficiency, anti-growth signal insensitivity, sustained angiogenesis, limitless replicative potential, tissue invasion, and metastasis.

Normal cells



Normal cells with basal → apical polarity. See cervical example **A**, which shows normal cells and spectrum of dysplasia, as discussed below.

Dysplasia



Abnormal proliferation of cells with loss of size, shape, and orientation (eg, koilocytic change, arrow in **A**). Compare vs hyperplasia (cells ↑ in number).

Carcinoma in situ/preinvasive

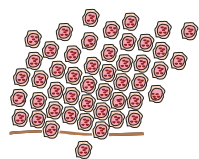


Neoplastic cells have not invaded the intact basement membrane.

↑ nuclear:cytoplasmic ratio and clumped chromatin.

Neoplastic cells encompass entire thickness.

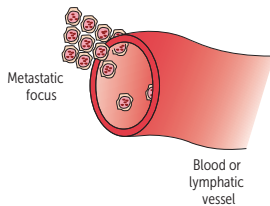
Invasive carcinoma



Cells have invaded basement membrane using collagenases and hydrolases (metalloproteinases).

Cell-cell contacts lost by inactivation of E-cadherin.

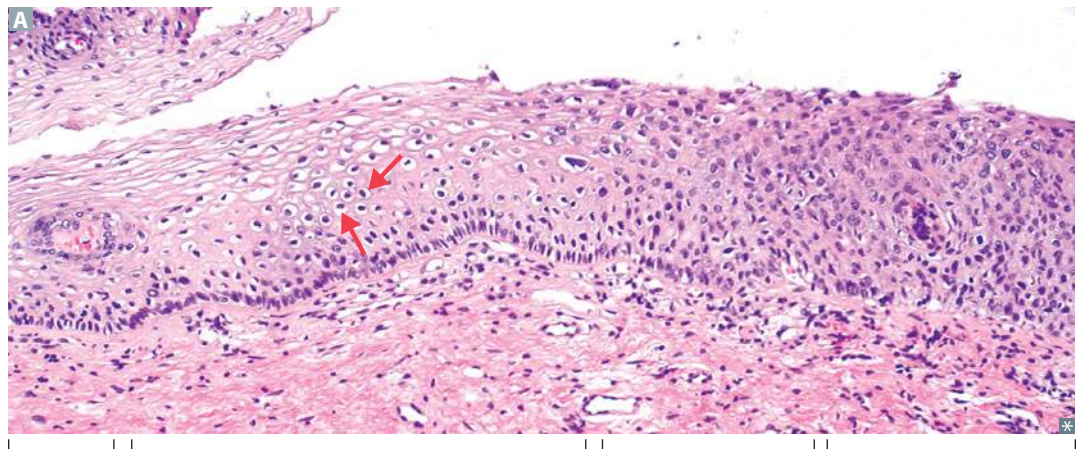
Metastasis



Spread to distant organ via lymphatics or blood.

“Seed and soil” theory of metastasis:

- Seed = tumor embolus.
- Soil = target organ is often the first-encountered capillary bed (eg, liver, lungs, bone, brain, etc).



Normal

Mild dysplasia

Moderate dysplasia

Severe dysplasia/
carcinoma in situ

Tumor grade vs stage

Grade	Degree of cellular differentiation and mitotic activity on histology. Range from low grade (well differentiated) to high grade (poorly differentiated, undifferentiated or anaplastic).	Stage generally has more prognostic value than grade (eg, a high-stage yet low-grade tumor is usually worse than a low-stage yet high-grade tumor).
Stage	Degree of localization/spread based on site and size of 1° lesion, spread to regional lymph nodes, presence of metastases. Based on clinical (c) or pathology (p) findings. Example: cT3N1M0	TNM staging system (S tage = S pread): T = T umor size/invasiveness N = N ode involvement M = M etastases Each TNM factor has independent prognostic value; N and M factors are often most important.

Tumor nomenclature

Carcinoma implies epithelial origin, whereas **sarcoma** denotes mesenchymal origin. Both terms imply malignancy.

Benign tumors are usually well differentiated, well demarcated, low mitotic activity, no metastasis, no necrosis.

Malignant tumors may show poor differentiation, erratic growth, local invasion, metastasis, and ↓ apoptosis. Upregulation of telomerase prevents chromosome shortening and cell death.

Terms for non-neoplastic malformations include hamartoma (disorganized overgrowth of tissues in their native location, eg, Peutz-Jeghers polyps) and choristoma (normal tissue in a foreign location, eg, gastric tissue located in distal ileum in Meckel diverticulum).

CELL TYPE	BENIGN	MALIGNANT
Epithelium	Adenoma, papilloma	Adenocarcinoma, papillary carcinoma
Mesenchyme		
Blood cells		Leukemia, lymphoma
Blood vessels	Hemangioma	Angiosarcoma
Smooth muscle	Leiomyoma	Leiomyosarcoma
Striated muscle	Rhabdomyoma	Rhabdomyosarcoma
Connective tissue	Fibroma	Fibrosarcoma
Bone	Osteoma	Osteosarcoma
Fat	Lipoma	Liposarcoma
Melanocyte	Nevus/mole	Melanoma

Cancer epidemiology

Skin cancer (basal > squamous >> melanoma) is the most common cancer (not included below).

	MEN	WOMEN	CHILDREN (AGE 0–14)	NOTES
Cancer incidence	1. Prostate 2. Lung 3. Colon/rectum	1. Breast 2. Lung 3. Colon/rectum	1. Leukemia 2. Brain and CNS 3. Neuroblastoma	Lung cancer incidence has dropped in men, but has not changed significantly in women.
Cancer mortality	1. Lung 2. Prostate 3. Colon/rectum	1. Lung 2. Breast 3. Colon/rectum	1. Leukemia 2. Brain and CNS 3. Neuroblastoma	Cancer is the 2nd leading cause of death in the United States (heart disease is 1st).

Paraneoplastic syndromes

MANIFESTATION	DESCRIPTION/MECHANISM	MOST COMMONLY ASSOCIATED CANCER(S)
Cutaneous		
Acanthosis nigricans	Hyperpigmented velvety plaques in axilla and neck	Gastric adenocarcinoma and other visceral malignancies (but more commonly associated with obesity and insulin resistance)
Sign of Leser-Trélat	Sudden onset of multiple seborrheic keratoses	GI adenocarcinomas and other visceral malignancies
Endocrine		
Hypercalcemia	PTHrP	Squamous cell carcinomas of lung, head, and neck; renal, bladder, breast, and ovarian carcinomas
	↑ 1,25-(OH) ₂ vitamin D ₃ (calcitriol)	Lymphoma
Cushing syndrome	↑ ACTH	Small cell lung cancer
Hyponatremia (SIADH)	↑ ADH	
Hematologic		
Polycythemia	↑ Erythropoietin	Renal cell carcinoma, hepatocellular carcinoma, hemangioblastoma, pheochromocytoma, leiomyoma
Pure red cell aplasia	Anemia with low reticulocytes	Thymoma
Good syndrome	Hypogammaglobulinemia	
Trousseau syndrome	Migratory superficial thrombophlebitis	Adenocarcinomas, especially pancreatic
Nonbacterial thrombotic (marantic) endocarditis	Deposition of sterile platelet thrombi on heart valves	
Neuromuscular		
Anti-NMDA receptor encephalitis	Psychiatric disturbance, memory deficits, seizures, dyskinesias, autonomic instability, language dysfunction	Ovarian teratoma
Opsoclonus-myoclonus ataxia syndrome	“Dancing eyes, dancing feet”	Neuroblastoma (children), small cell lung cancer (adults)
Paraneoplastic cerebellar degeneration	Antibodies against antigens in Purkinje cells	Small cell lung cancer (anti-Hu), gynecologic and breast cancers (anti-Yo), and Hodgkin lymphoma (anti-Tr)
Paraneoplastic encephalomyelitis	Antibodies against Hu antigens in neurons	Small cell lung cancer
Lambert-Eaton myasthenic syndrome	Antibodies against presynaptic (P/Q-type) Ca ²⁺ channels at NMJ	Small cell lung cancer
Myasthenia gravis	Antibodies against postsynaptic ACh receptors at NMJ	Thymoma

OncogenesGain of function → ↑ cancer risk. Need damage to only **one** allele of an **oncogene**.

GENE	GENE PRODUCT	ASSOCIATED NEOPLASM
ALK	Receptor tyrosine kinase	Lung adenocarcinoma
BCR-ABL	Tyrosine kinase	CML, ALL
BCL-2	Antiapoptotic molecule (inhibits apoptosis)	Follicular and diffuse large B cell lymphomas
BRAF	Serine/threonine kinase	Melanoma, non-Hodgkin lymphoma, papillary thyroid carcinoma
c-KIT	Cytokine receptor	Gastrointestinal stromal tumor (GIST)
c-MYC	Transcription factor	Burkitt lymphoma
HER2/neu (c-erbB2)	Receptor tyrosine kinase	Breast and gastric carcinomas
JAK2	Tyrosine kinase	Chronic myeloproliferative disorders
KRAS	GTPase	Colon cancer, lung cancer, pancreatic cancer
MYCL1	Transcription factor	Lung tumor
MYCN	Transcription factor	Neuroblastoma
RET	Receptor tyrosine kinase	MEN 2A and 2B, medullary thyroid cancer

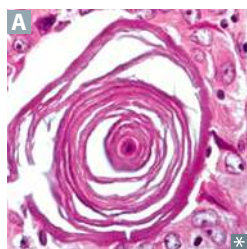
Tumor suppressor genesLoss of function → ↑ cancer risk; both (**two**) alleles of a **tumor suppressor gene** must be lost for expression of disease.

GENE	GENE PRODUCT	ASSOCIATED CONDITION
APC	Negative regulator of β -catenin/WNT pathway	Colorectal cancer (associated with FAP)
BRCA1/BRCA2	DNA repair protein	Breast , ovarian, and pancreatic cancer
CDKN2A	p16, blocks $G_1 \rightarrow S$ phase	Melanoma, pancreatic cancer
DCC	DCC —Deleted in Colon Cancer	Colon cancer
DPC4/SMAD4	DPC —Deleted in Pancreatic Cancer	Pancreatic cancer
MEN1	Menin	MEN 1
NF1	Neurofibromin (Ras GTPase activating protein)	Neurofibromatosis type 1
NF2	Merlin (schwannomin) protein	Neurofibromatosis type 2
PTEN	Tyrosine phosphatase of PIP_3 (eg, protein kinase B [AKT] activation)	Breast cancer, prostate cancer, endometrial cancer
Rb	Inhibits E2F; blocks $G_1 \rightarrow S$ phase	Retinoblastoma , osteosarcoma
TP53	p53, activates p21, blocks $G_1 \rightarrow S$ phase	Most human cancers, Li-Fraumeni syndrome (multiple malignancies at early age, aka, SBLA cancer syndrome: Sarcoma , Breast , Leukemia , Adrenal gland)
TSC1	Hamartin protein	Tuberous sclerosis
TSC2	Tuberin protein	Tuberous sclerosis
VHL	Inhibits hypoxia inducible factor 1 α	von Hippel-Lindau disease
WT1	Transcription factor that regulates urogenital development	Wilms tumor (nephroblastoma)

Oncogenic microbes	Microbe	Associated cancer
	EBV	Burkitt lymphoma, Hodgkin lymphoma, nasopharyngeal carcinoma, 1° CNS lymphoma (in immunocompromised patients)
	HBV, HCV	Hepatocellular carcinoma
	HHV-8	Kaposi sarcoma
	HPV	Cervical and penile/anal carcinoma (types 16, 18), head and neck cancer
	<i>H pylori</i>	Gastric adenocarcinoma and MALT lymphoma
	HTLV-1	Adult T-cell leukemia/lymphoma
	Liver fluke (<i>Clonorchis sinensis</i>)	Cholangiocarcinoma
	<i>Schistosoma haematobium</i>	Bladder cancer (squamous cell)

Carcinogens

TOXIN	ORGAN	IMPACT
Aflatoxins (<i>Aspergillus</i>)	Liver	Hepatocellular carcinoma
Alkylating agents	Blood	Leukemia/lymphoma
Aromatic amines (eg, benzidine, 2-naphthylamine)	Bladder	Transitional cell carcinoma
Arsenic	Liver	Angiosarcoma
	Lung	Lung cancer
	Skin	Squamous cell carcinoma
Asbestos	Lung	Bronchogenic carcinoma > mesothelioma
Carbon tetrachloride	Liver	Centrilobular necrosis, fatty change
Cigarette smoke	Bladder	Transitional cell carcinoma
	Cervix	Cervical carcinoma
	Esophagus	Squamous cell carcinoma/adenocarcinoma
	Kidney	Renal cell carcinoma
	Larynx	Squamous cell carcinoma
	Lung	Squamous cell and small cell carcinoma
	Pancreas	Pancreatic adenocarcinoma
Ethanol	Esophagus	Squamous cell carcinoma
	Liver	Hepatocellular carcinoma
Ionizing radiation	Thyroid	Papillary thyroid carcinoma
Nitrosamines (smoked foods)	Stomach	Gastric cancer
Radon	Lung	Lung cancer (2nd leading cause after cigarette smoke)
Vinyl chloride	Liver	Angiosarcoma

Psammoma bodies

Laminated, concentric spherules with dystrophic calcification **A**, **PSaMM**oma bodies are seen in:

- **P**apillary carcinoma of thyroid
- **S**erous papillary cystadenocarcinoma of ovary
- **M**eningioma
- Malignant **M**esothelioma

Serum tumor markers

Tumor markers should not be used as the 1° tool for cancer diagnosis or screening. They may be used to monitor tumor recurrence and response to therapy, but definitive diagnosis is made via biopsy.

MARKER	ASSOCIATED CANCER	NOTES
Alkaline phosphatase	Metastases to bone or liver, Paget disease of bone, seminoma (placental ALP).	Must exclude hepatic origin by checking LFTs and GGT levels.
α-fetoprotein	Hepatocellular carcinoma, hepatoblastoma, yolk sac (endodermal sinus) tumor, mixed germ cell tumor.	Normally made by fetus. Transiently elevated in pregnancy. High levels associated with neural tube and abdominal wall defects, low levels associated with Down syndrome.
β-hCG	H ydattidiform moles and C horiocarcinomas (G estational trophoblastic disease), testicular cancer, mixed germ cell tumor.	Produced by syncytiotrophoblasts of the placenta.
CA 15-3/CA 27-29	Breast cancer.	
CA 19-9	Pancreatic adenocarcinoma.	
CA 125	Ovarian cancer.	
Calcitonin	Medullary thyroid carcinoma (alone and in MEN2A, MEN2B).	
CEA	Major associations: colorectal and pancreatic cancers. Minor associations: gastric, breast, and medullary thyroid carcinomas.	C arcinoembryonic a ntigen. Very nonspecific.
Chromogranin	Neuroendocrine tumors.	
PSA	Prostate cancer.	P rostate-specific a ntigen. Can also be elevated in BPH and prostatitis. Questionable risk/benefit for screening. Surveillance marker for recurrent disease after prostatectomy.

P-glycoprotein

Also known as multidrug resistance protein 1 (MDR1). Classically seen in adrenocortical carcinoma but also expressed by other cancer cells (eg, colon, liver). Used to pump out toxins, including chemotherapeutic agents (one mechanism of ↓ responsiveness or resistance to chemotherapy over time).

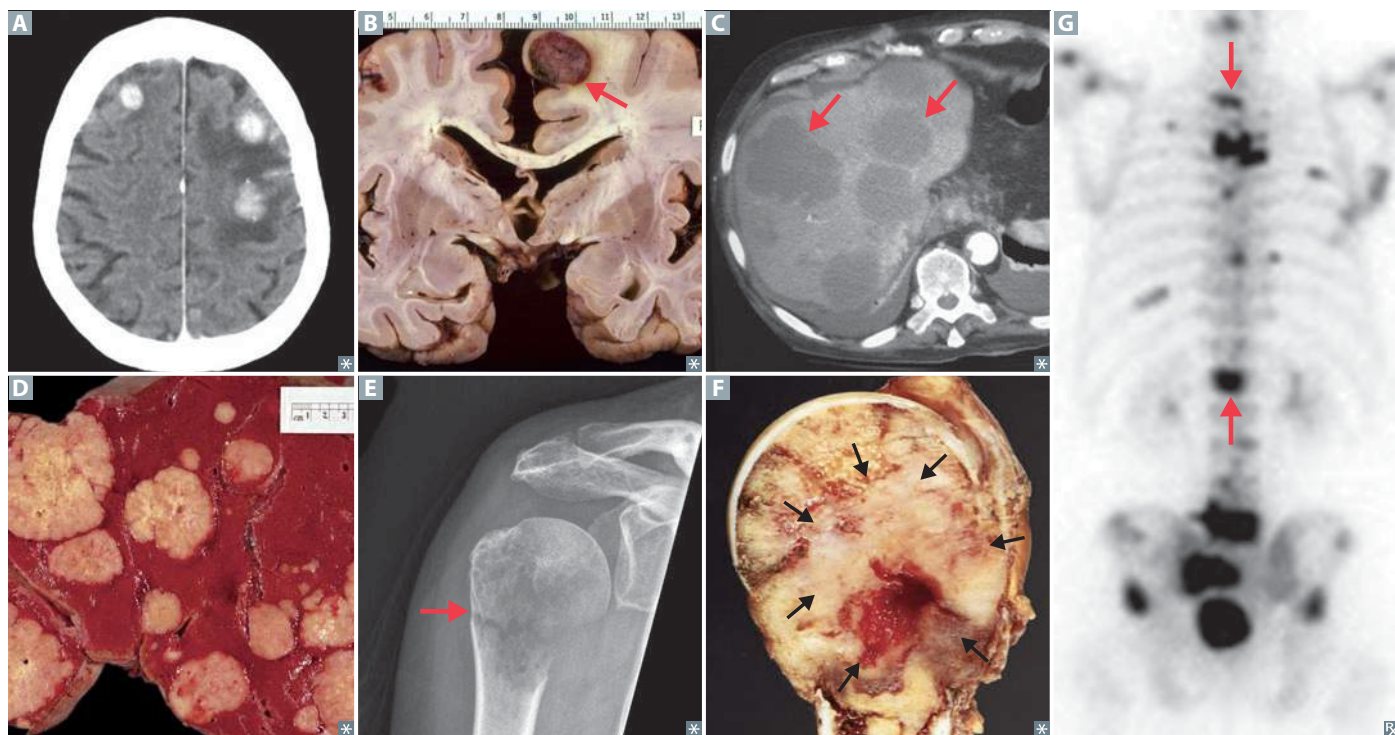
Cachexia

Weight loss, muscle atrophy, and fatigue that occur in chronic disease (eg, cancer, AIDS, heart failure, COPD). Mediated by TNF, IFN- γ , IL-1, and IL-6.

Common metastases

Most sarcomas spread hematogenously; most carcinomas spread via lymphatics. However, **Four Carcinomas Route Hematogenously: Follicular thyroid carcinoma, Choriocarcinoma, Renal cell carcinoma, and Hepatocellular carcinoma.**

SITE OF METASTASIS	1° TUMOR	NOTES
Brain	Lung > breast > melanoma, colon, kidney.	50% of brain tumors are from metastases A B . Commonly seen as multiple well-circumscribed tumors at gray/white matter junction.
Liver	Colon >> stomach > pancreas.	Liver C D and lung are the most common sites of metastasis after the regional lymph nodes.
Bone	Prostate, breast > lung, thyroid, kidney.	Bone metastasis E F >> 1° bone tumors (eg, multiple myeloma, lytic). Common mets to bone: breast (mixed), lung (lytic), thyroid (lytic), kidney (lytic), prostate (blastic). Predilection for axial skeleton G .



▶ NOTES